

Basic course

KNX Topology

KNX Association

Table of Contents

1	<i>KNX TP Topology - Overall TP view [E]</i>	4
2	<i>Topology – Line segment [E]</i>	5
3	<i>Topology – Area</i>	6
4	<i>Topology - Several areas (Network)</i>	7
5	<i>Individual address in the TP topology [E]</i>	8
6	<i>KNX Couplers with a gate function in KNX</i>	10
7	<i>KNX Coupler - Block diagram overview</i>	11
8	<i>Couplers - Fields of application</i>	12
8.1	<i>Introduction of segment couplers in KNX</i>	13
8.1.1	<i>Segment couplers in ETS6:</i>	14
9	<i>Connecting several lines</i>	18
10	<i>Practical example for explanation of functionality [E]</i>	19
11	<i>Internal line telegram [E]</i>	20
12	<i>Line-crossing telegram</i>	21
13	<i>Area-crossing telegram</i>	22
14	<i>Hop count in KNX couplers</i>	23
15	<i>TP Topology – Building structure</i>	24
16	<i>Backbone- /line coupler classical structure</i>	25
17	<i>KNX internal and external interfaces</i>	26
18	<i>KNX IP network for higher telegram rates</i>	28
19	<i>Line couplers replaced by KNXnet/IP routers</i>	30
20	<i>Limits to using KNXnet/IP routers</i>	31

Preface

This training document is created for the Basic course. Sections with a heading marked [E] contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does not have [E], it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e. S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

In this chapter, the following abbreviations are used:

BC = Backbone coupler

LC = Line coupler

DVC = Bus device

LR = Line repeater

PS = Power supply with choke

S = Brightness sensor

HC = Hop count (used to be referred to as Routing counter (RC))

SC = Segment coupler

1 KNX TP Topology - Overall TP view [E]

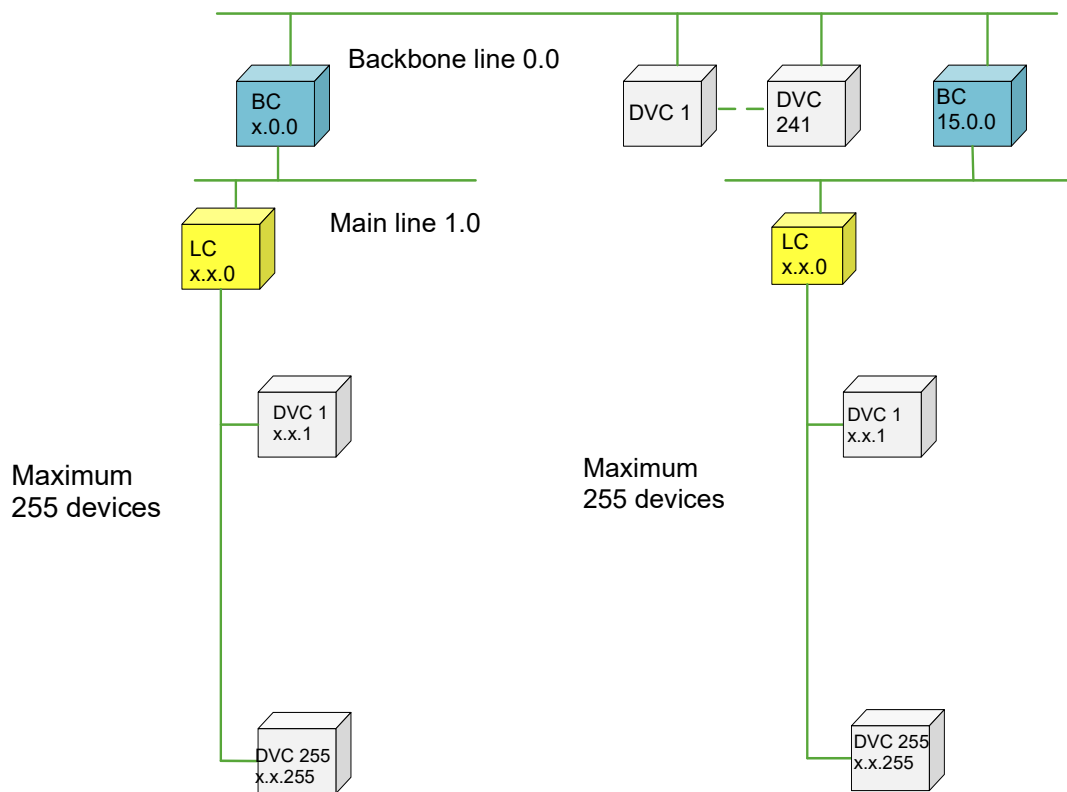


Figure 1: Maximum topological size of a KNX TP installation

The maximum topological size of a KNX TP installation is shown in the figure above. The following sections will explain the details of a KNX TP topology step by step. Throughout this explanation and other parts of this course, the following terms are used:

Primary line (North side):

- Refers to the upper segment of a line - or backbone coupler.
- It can be different depending on the type of coupler:
 - **For a line coupler** → The TP main line is the primary line.
 - **For an IP router** → The IP side is the primary line.

Secondary line (South side):

- Refers to the lower segment of a line - or backbone coupler.
- Sometimes called the sub-line.
- Can also differ based on the **type of coupler**:
 - **For a line coupler**: The actual TP line is the secondary line.
 - **For an IP router**: The TP side is the secondary line.

2 Topology – Line segment [E]

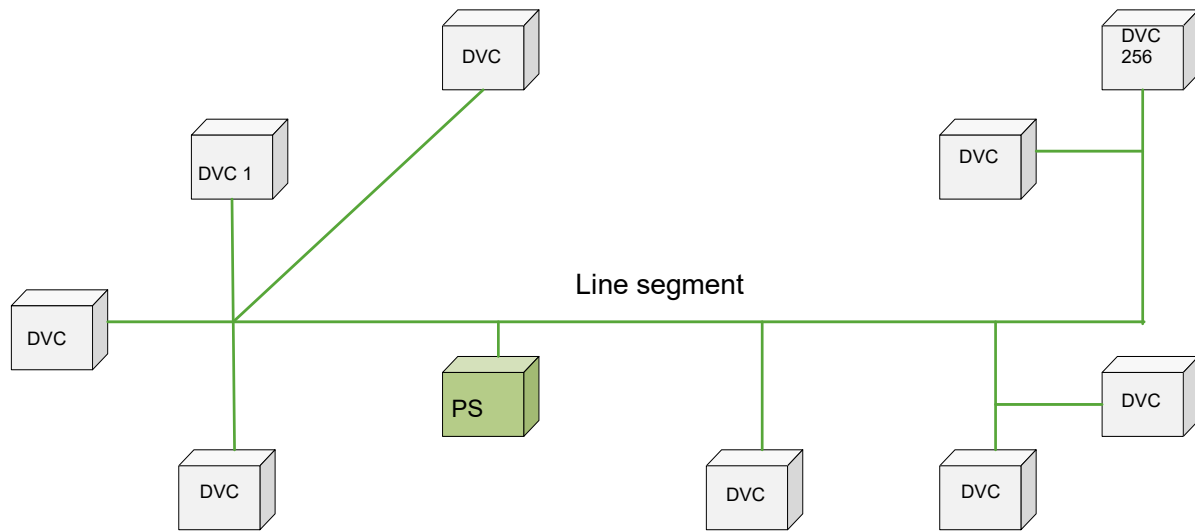


Figure 2: Topology - line segment

A line segment is the smallest unit in a KNX TP topology. The key characteristics of a KNX TP line segment are:

- Maximum 256 devices per line segment.
- Devices communicate via telegrams and can exchange data with any other device in the same line segment.
- Each line segment requires its own power supply to ensure all devices function correctly.
- The actual number of devices that can be supported depends on:
 - The selected power supply unit.
 - The power consumption of each device.

Allowed bus cable structures:

The bus cable can be branched anywhere and supports the following topologies:

- Star topology: Devices are connected in a star-shaped layout.
- Line topology (Daisy Chain): Devices are connected one after another in a straight line.
- Tree topology: A mix of star and line structures, optimizing cable usage.

Loop/Ring structures are NOT allowed because they can cause network conflicts.

3 Topology – Area

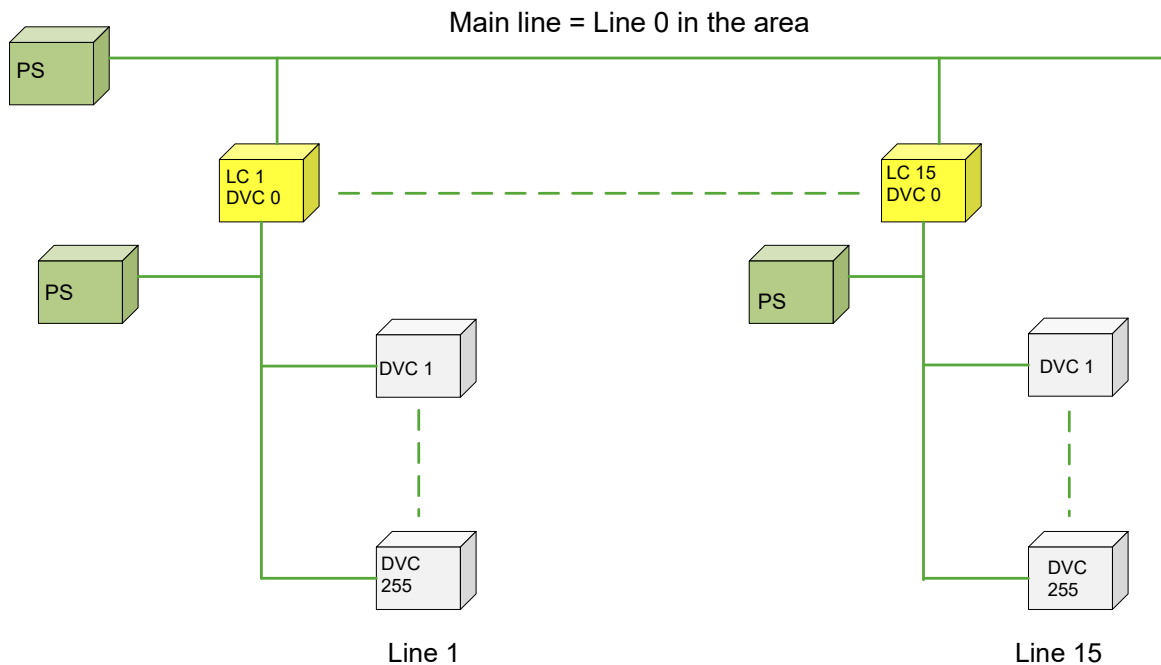


Figure 3: Topology – area

If a single line segment is not enough to install all devices, up to **15 lines** can be connected to a main line using line couplers. This structure is called an **area**.

Line couplers are considered part of the secondary line, but their power consumption from the main line reduces the maximum number of devices that can be installed on the main line. This means that up to 256 devices can be installed on the main line. However, considering the 16 line – and backbone couplers, not all available individual addresses on the main line can be used simultaneously.

Every secondary line segment and the main line must have a separate KNX power supply unit. With up to 15 secondary lines per area, a KNX TP installation can support over 4,000 devices. This scalability makes KNX TP suitable for large buildings and complex automation projects.

Note: Line repeaters may not be used on the main line.

4 Topology - Several areas (Network)

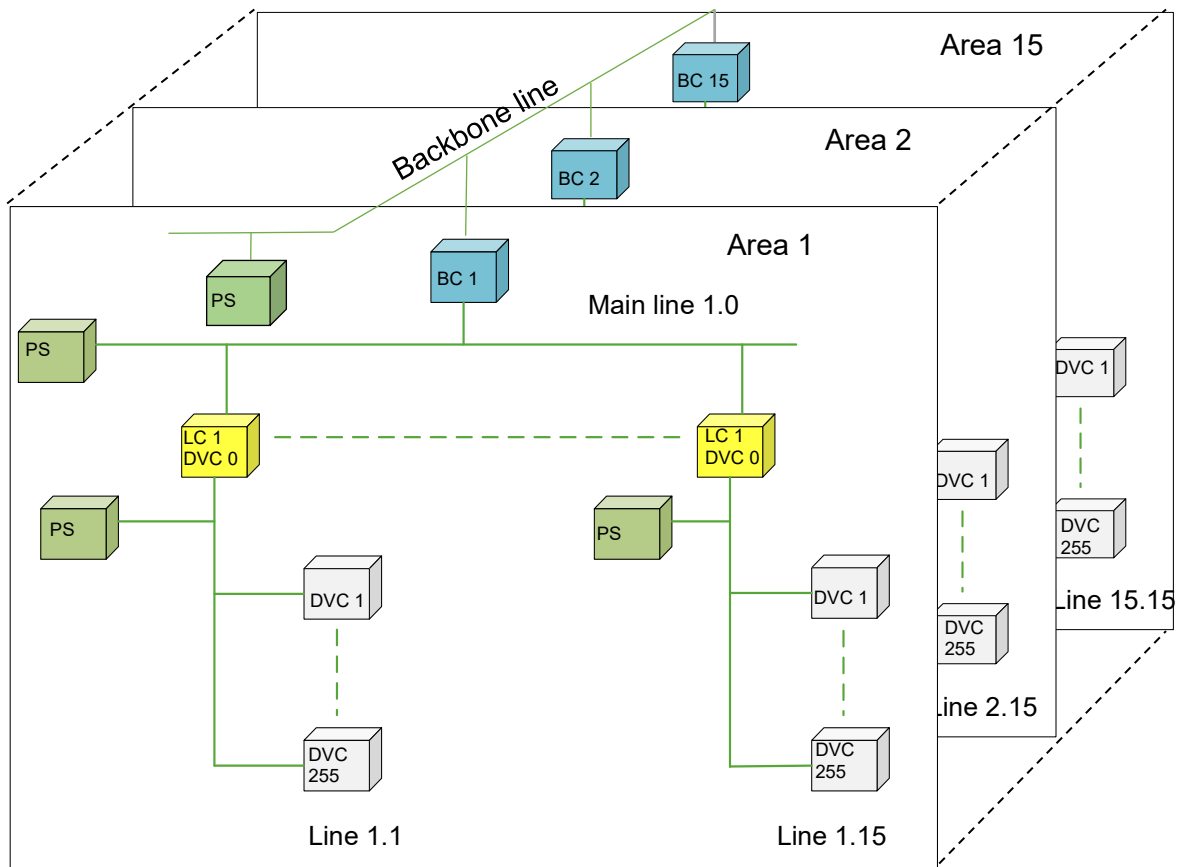


Figure 4: Topology - several areas

When more devices are needed in a KNX TP installation, the system can be expanded by adding a backbone line using backbone couplers.

Just like line couplers, backbone couplers are considered part of the secondary line, the main line in this case, but their power consumption from the backbone line reduces the maximum number of devices that can be installed on the backbone line.

The backbone line must have a separate KNX power supply unit. By using lines, areas, and a backbone structure, a KNX TP network can scale up to more than 61,000 devices.

Note: Line repeaters may not be used on the backbone line.

5 Individual address in the TP topology [E]

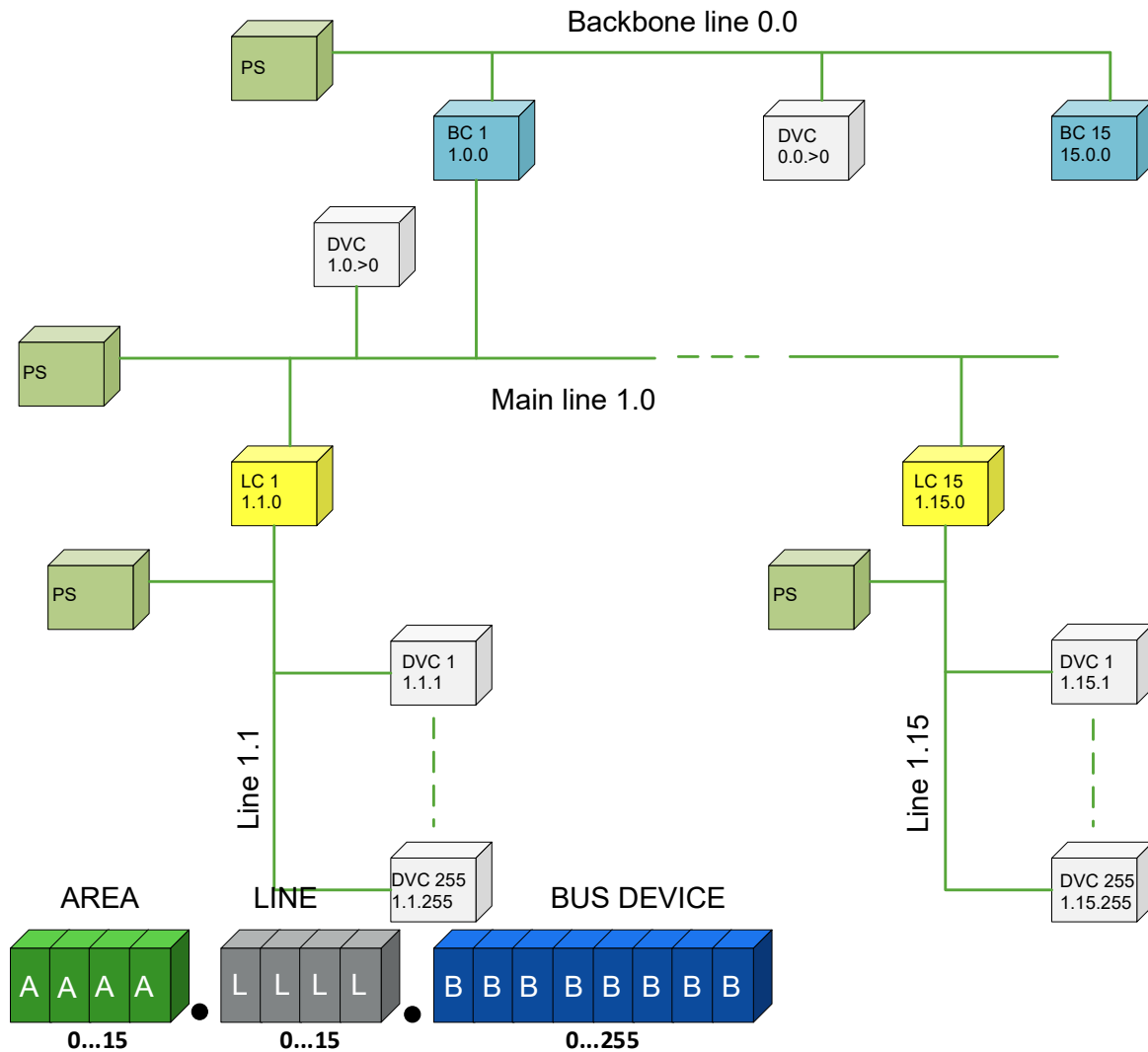


Figure 5: Individual address

Each KNX TP bus device is assigned a unique individual address to clearly identify its location within the topology. The address format is: A.L.B (Area.Line.Bus device)

	Range	Explanation
A =	1...15	Addresses the areas 1...15.
A =	0	Addresses the bus devices on the backbone line.
L =	1...15	Addresses the lines 1...15 in the areas defined by A.
L =	0	Addresses the main line of the respective area.
B =	1...255	Addresses the bus devices on the line defined by L.
B =	0	Addresses the line coupler in the respective line.

When a new KNX bus device is first powered on, it usually has the default individual address **15.15.255**. After an unload by ETS, a bus device always has the default individual address 15.15.255 with the exception for couplers and IP routers (15.15.0). This means the device is unloaded and has not yet been assigned to a specific location in the KNX topology. All new KNX devices are delivered from the factory with this address until they are configured in ETS.

6 KNX Couplers with a gate function in KNX

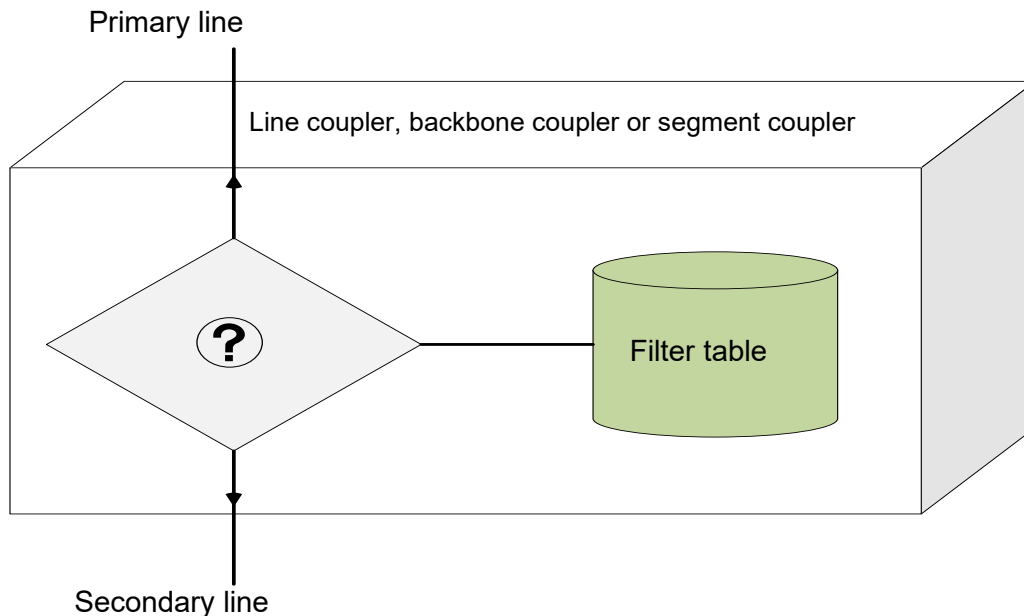


Figure 6: KNX couplers with a gate function

Couplers (line -, backbone - and segment couplers) manage communication between different KNX lines by filtering telegrams.

Each coupler has a **gate** function, controlling which telegrams pass through. When a correct individual address is assigned to a coupler, ETS automatically creates a **filter table** for it during the planning and design stage. This filter table contains **only group addresses** that are used by bus devices **on both sides of the coupler**.

The coupler only forwards group addresses that are listed in its filter table. Only necessary telegrams are forwarded, ensuring that each line functions independently. This reduces unnecessary traffic and improves network efficiency.

The yellow LEDs flicker when the coupler receives and processes telegrams from the respective line.

Note: Line repeaters do not have a filter table and pass on all telegrams without filtering (See Couplers - Fields of application).

7 KNX Coupler - Block diagram overview

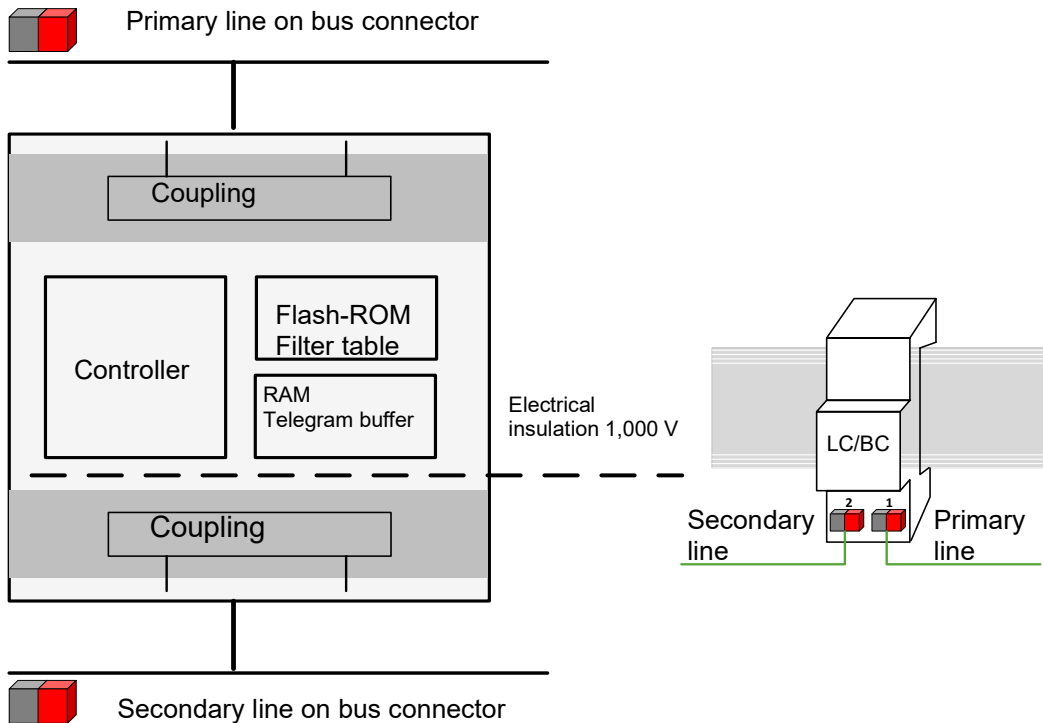


Figure 7: Line coupler: Block Diagram

A KNX coupler is a (DIN rail-mounted) device that connects a primary line to a secondary line, ensuring controlled communication between them. The primary line and the secondary line are connected with standardized **bus connectors** to the coupler.

The coupler can be programmed from both the primary and secondary line, allowing flexibility in system configuration.

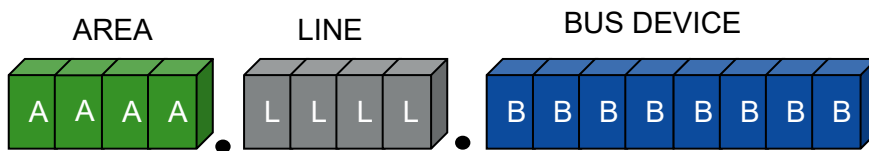
A coupler is powered from the primary line. For its intended operation as a “coupler” it also needs to be connected to a secondary line with its own power supply unit.

The fact that it is powered from the primary line gives the opportunity to detect and report a secondary line power failure.

The couplers have a Flash ROM memory and provide galvanic isolation between the primary and secondary line.

8 Couplers - Fields of application

Individual address



The coupler is used

			As	For
>0	=0	=0	Backbone coupler	Backbone line/Main line
>0	>0	=0	Line coupler	Main line/Secondary line
>0	>0	>0	Line repeater	Line extension
>=0	>0	>0	Segment coupler	Extra segment

Figure 8: Coupler: fields of application

Backbone couplers, line couplers and line repeaters are physically identical devices but serve different functions based on their location and assigned individual address.

Types of couplers & their functions:

Coupler type	Connection between	Function
Backbone coupler	Backbone line <-> Main line	Connect up to 15 areas via the backbone.
Line coupler	Main Line <-> Secondary line	Connect up to 15 lines to the main line in an area.
Segment coupler	Main segment and additional subsegments	For creating extra subsegment(s) with filtering functionality
Line repeater (can be substituted by segment couplers)	Main segment and additional subsegments	Expands a line segment by up to 3 more segments , should the available cable length of 1,000 m not be sufficient for mounting the maximum number of 256 KNX bus devices ¹ . They do not have filtering function.

Backbone couplers, line couplers and segment couplers only forward line-crossing telegrams (based on the filter tables in ETS). The line repeater does not filter any telegrams and forwards all telegrams. The coupler's role (BC, LC, or LR) is **determined by its individual address** in the KNX topology, e.g.,

- Address 1.1.0: Line coupler for line 1 in area 1.
- Address 8.0.0: Backbone coupler for area 8.

¹ If you can install all 256 TP devices within a cable length of 1,000 m, then there is no need to install any Line Repeaters. However, if your installation requires a cable length exceeding 1,000 m for these 256 devices, you must install a Line Repeater for every additional line segment of 1,000 m. The maximum allowed configuration is 3 Line Repeaters, resulting in a total maximum cable length of 4,000 m.

Importance of correct addressing:

- KNX couplers recognize the devices within their area or line using their own individual address.
- See the following example when a specific bus device will be downloaded by means of ETS:
 - A telegram addressed to device 1.2.20 is sent.
 - A line coupler with address 1.3.0 will **NOT** forward the telegram because 1.2.20 is not in its line.
 - Hence, devices must be mounted in the correct line and must be topologically assigned the correct individual address.

8.1 Introduction of segment couplers in KNX

Segment couplers were introduced in ETS6 to improve flexibility, filtering, and integration of different KNX media types (e.g., TP and RF).

Why were segment couplers introduced?

- To intelligently expand existing TP lines with RF (or additional TP) devices.
- To eliminate the need to install media couplers on the main line (see hybrid network in the RF section of the Basic course).
- Because RF media couplers could not be integrated if there was only a single TP line (e.g., 1.1) due to the lack of a main line.
- Segment couplers now enable the branching of a TP line to include RF (or TP) devices.
- Because they enable telegrams to be filtered between segments.
- Because TP line repeaters forward all data traffic, even to devices that do not need it.
- Because the telegram filtering of segment couplers significantly improves the efficiency of the network.

8.1.1 Segment couplers in ETS6:

From ETS6 onwards, the Topology panel and functionality have been enhanced to improve the management of segments and media types in KNX installations.

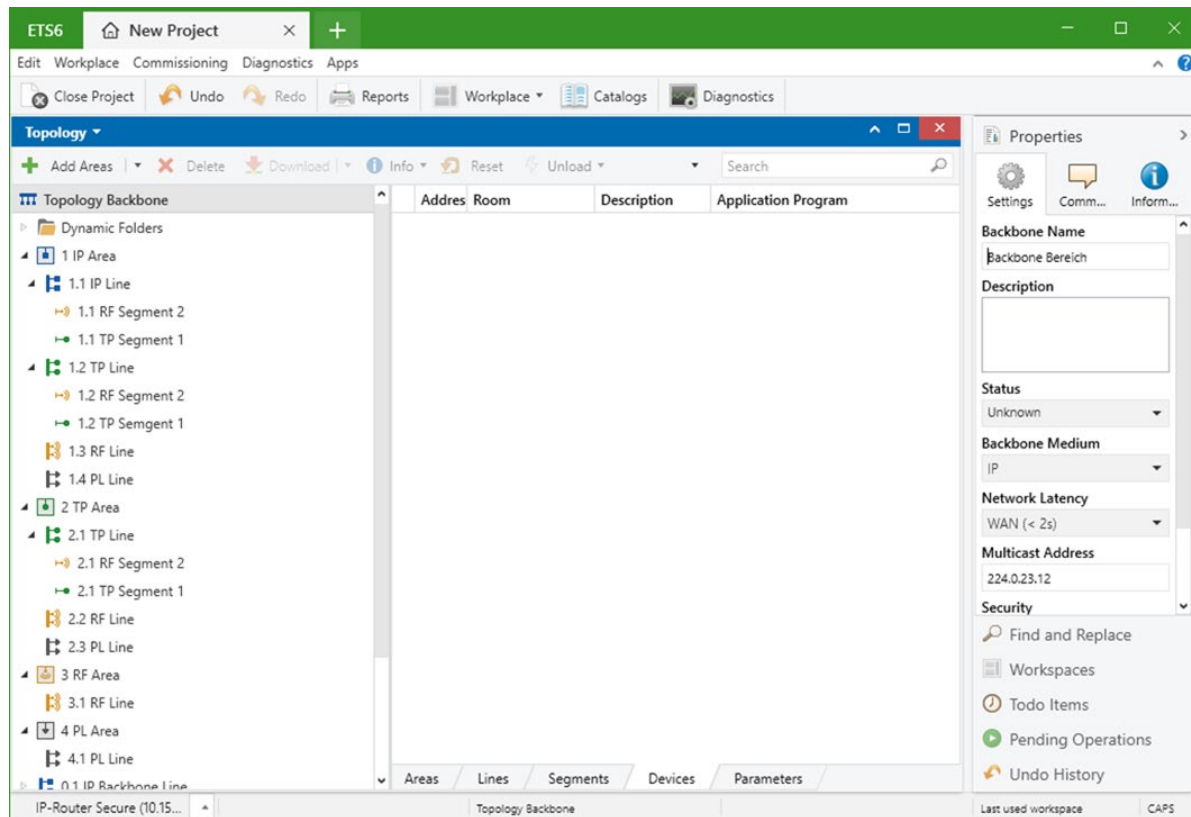


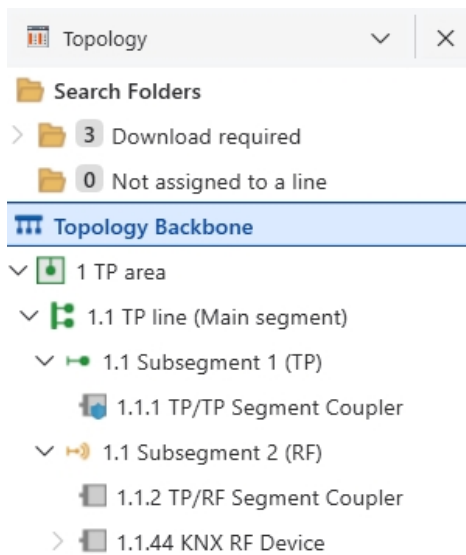
Figure 9: ETS Topology panel with segment couplers

There is defined color coding for the different media types:

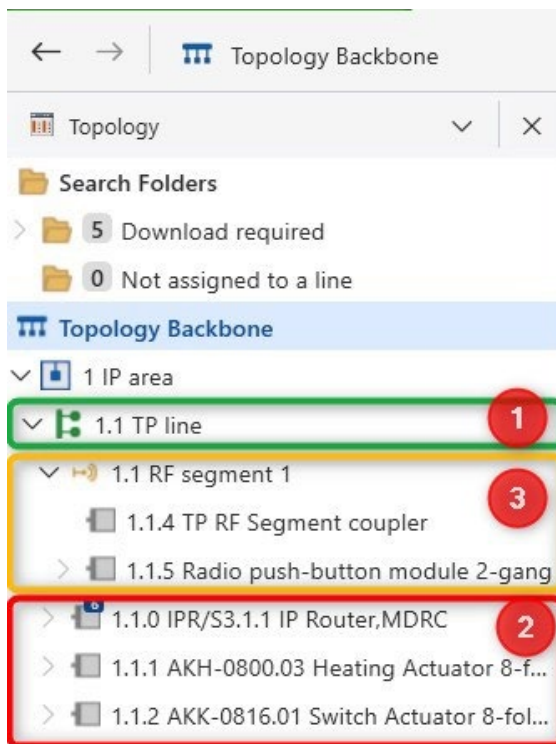
- Green: TP (Twisted Pair).
- Orange: RF (Radio Frequency).
- Blue: IP (Internet Protocol).

This visual distinction makes it easier to identify and configure different KNX segments.

- When selecting a line in ETS6, a dedicated tab displays all segments related to that line. This provides a clear overview of all devices and segments. In the Device Info of the segment coupler, it is now possible to preview:
 - Group address filter table.
 - Individual address filter table.
- To every KNX line (main segment), one can add multiple (TP or RF) subsegments.
- Subsegments are connected to the main segment via a segment coupler.
- Subsegments can have different media types (e.g., RF subsegment connected to TP). This means that an RF device with individual address 1.1.44 can be mounted in an RF subsegment behind a TP/RF segment coupler which itself is connected to the main TP segment.

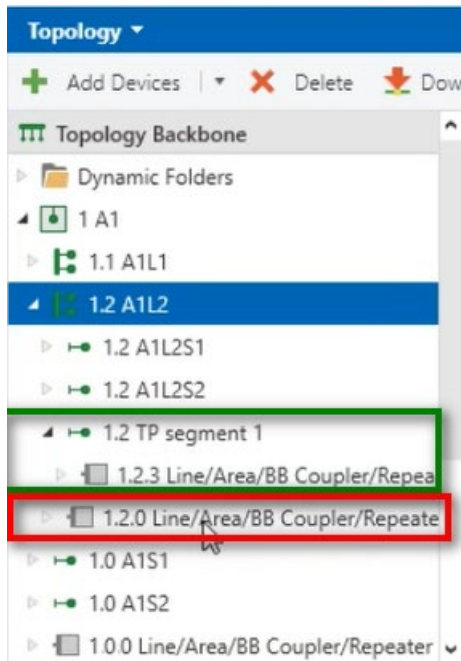


This figure illustrates a TP area with one TP line and two subsegments, one with medium TP and another one with medium RF. Next to one segment coupler, it is possible to add any number of additional devices.



This figure illustrates a TP line (Green) containing:

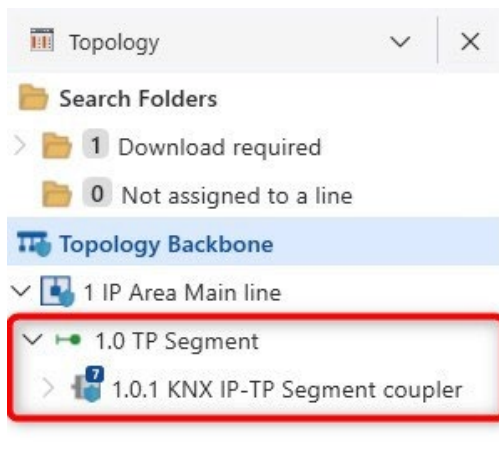
- TP devices (Red) connected to the TP main segment.
- An RF segment (Orange) with a TP/RF segment coupler and multiple RF devices.



This figure illustrates a TP area with several TP lines. One of these lines, line 1.2, includes a line coupler (1.2.0) and multiple segments.

When a coupler with repeater functionality is added to this TP line (using Drag and Drop in ETS6):

- A new segment is automatically created.
- The repeater is assigned to that segment (Green).



This figure illustrates a TP segment directly connected to the IP backbone, with an IP/TP segment coupler (Red) added to it.

When devices are placed within this segment, the entire segment can be copied and replicated using the “Paste Special” function (copying it multiple times) within the backbone. This approach enables the implementation of the “hotel scenario”.

Note: Device addresses are flexible:

- Devices do not need to follow strict address ranges per segment.
- Example:
 - Devices 1.1.77 and 1.1.90 can be in segment 1.
 - Device 1.1.85 can be in segment 2.

Remember:

- The total number of devices per line is still limited to 256, including all segment couplers.
- Each TP segment (main and subsegments) must have its own power supply to provide power to devices in that segment.

Address filtering in segment couplers:

Group addresses (for normal operation):

- Segment couplers like line - and backbone couplers have a filter table for group-addressed telegrams.
- ETS automatically manages this filter table based on which group addresses are linked to which devices in the segment.

Individual addresses (for downloading the devices via ETS):

- Segment couplers know which devices with which individual address are in their segment.
- While backbone and line couplers determine the device location based on their area and line address, segment couplers get this information from ETS (filter table for individual addresses).
- Each time you add a new device or change the address of a bus device in a subsegment, the segment coupler needs to be downloaded again in ETS.
- If you temporarily connect an interface to a segment (e.g., for diagnostic purposes), the ETS also temporarily adds its own address to the individual addresses that are to be (temporarily) filtered in the segment coupler.

9 Connecting several lines

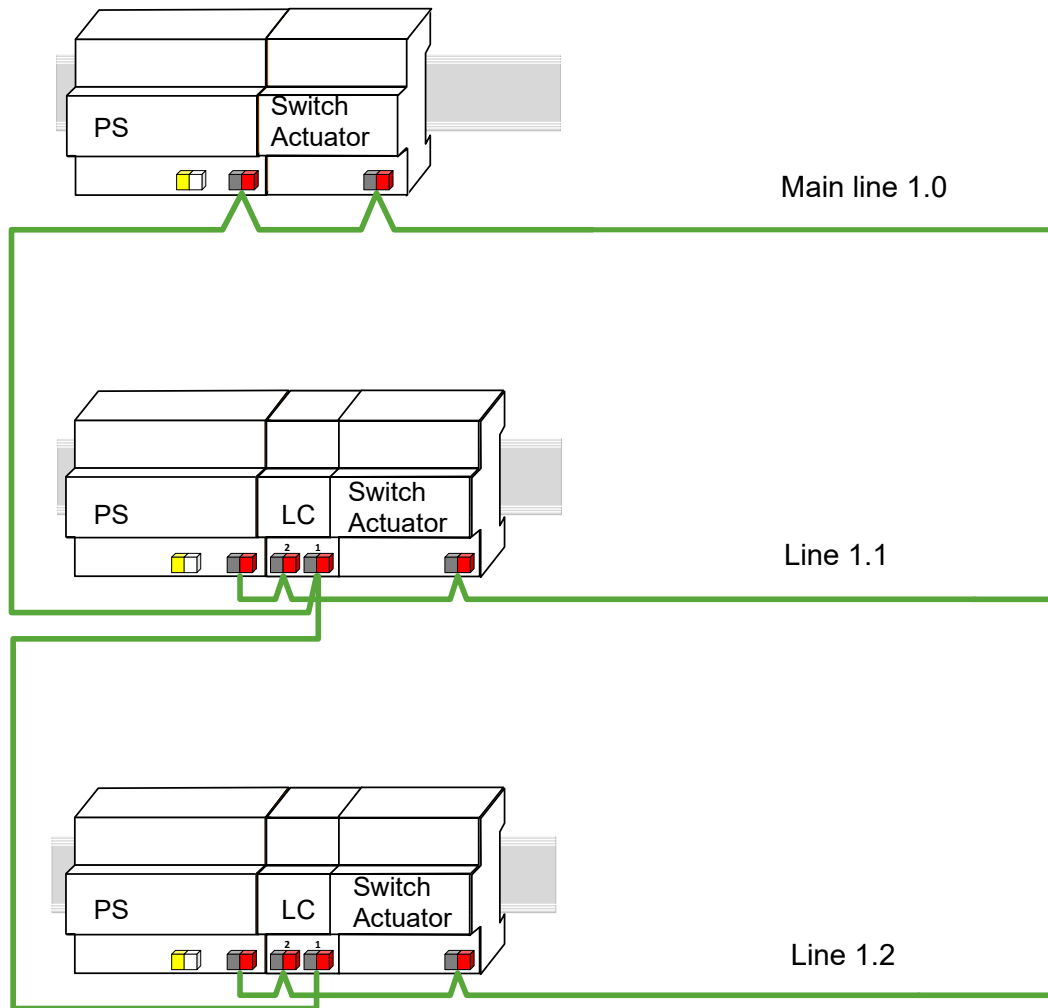


Figure 10: connecting several lines (example for area 1)

When multiple lines are used in a KNX TP installation, each line and line segment must have its own power supply unit and choke to ensure stable operation.

Both lines, the secondary line (e.g., line 1.1) as well as the primary line (main line 1.0) are connected to a line coupler via standardized bus connectors.

10 Practical example for explanation of functionality [E]

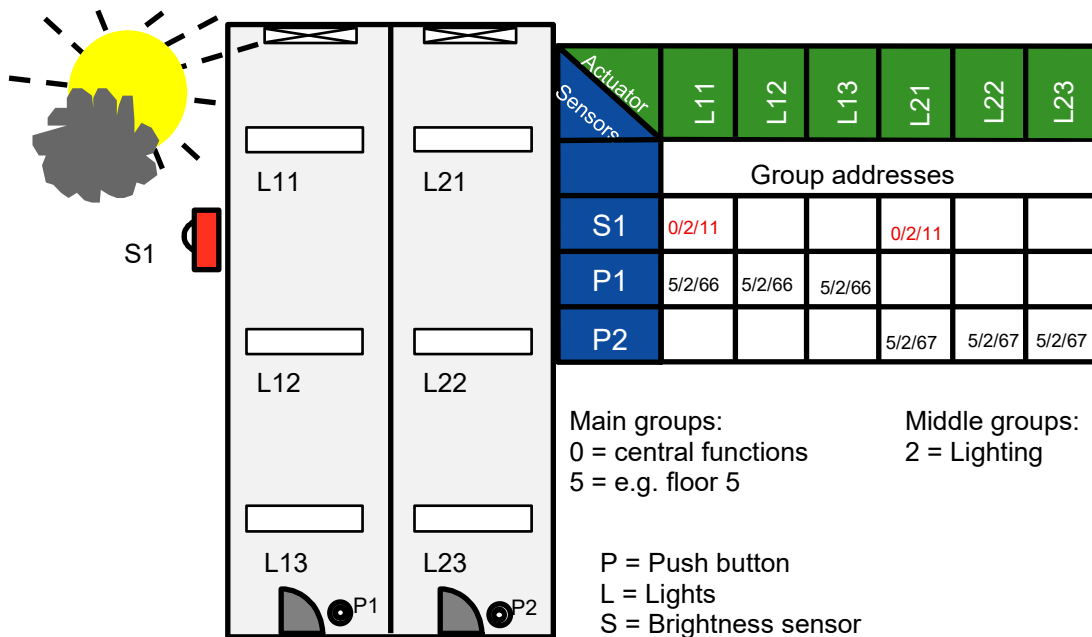


Figure 11: Practical example

The push button P1 controls lights L11, L12 and L13. During configuration in ETS, group address 5/2/66 is assigned to P1. The same group address is also assigned to the actuators controlling the lamps (L11, L12 and L13).

The push button P2 controls lights L21, L22 and L23. During configuration in ETS, the group address 5/2/67 is assigned to P2. Again, the same group address is assigned to the actuators controlling the lamps (L21, L22 and L23).

The brightness sensor S1 controls the window lights (L11 and L21). Group address 0/2/11 is therefore assigned to the sensor as well as to the actuators controlling the window lights (L11 and L21).

The window lights can therefore be switched via the push button (P1 and P2) as well as the brightness sensor (S1).

11 Internal line telegram [E]

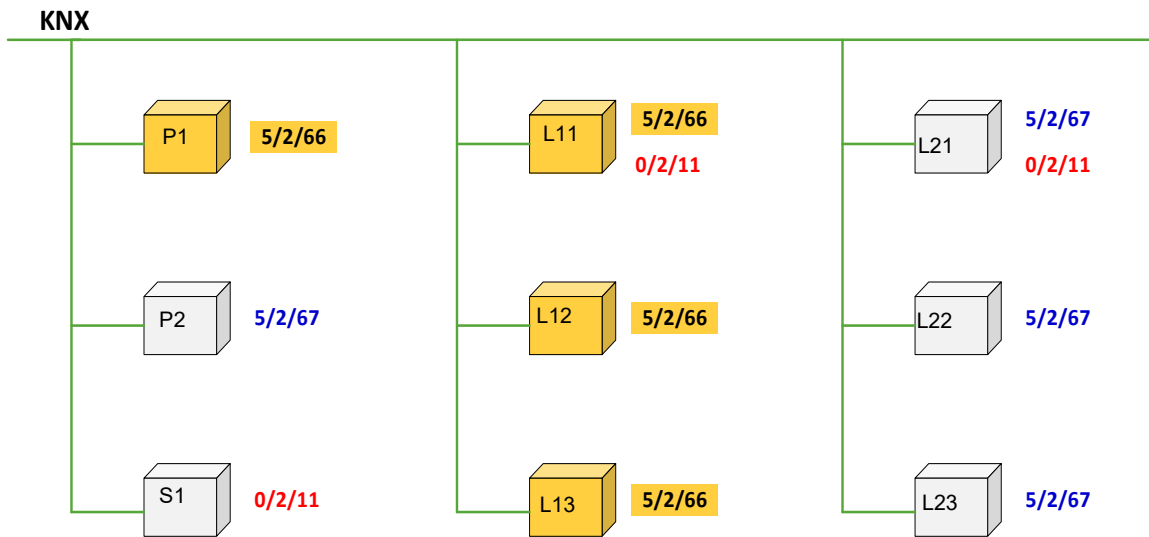


Figure 12: Internal line telegram

Pressing push button P1 sends a telegram with the group address 5/2/66.

All bus devices receive and analyze this telegram, but only the actuators of L11, L12, and L13 recognize this group address and execute the command.

If the brightness sensor (S1) sends the group address 0/2/11, all bus devices receive and analyze this telegram, but only the actuators of L11 and L21 (window lights) execute the command.

12 Line-crossing telegram

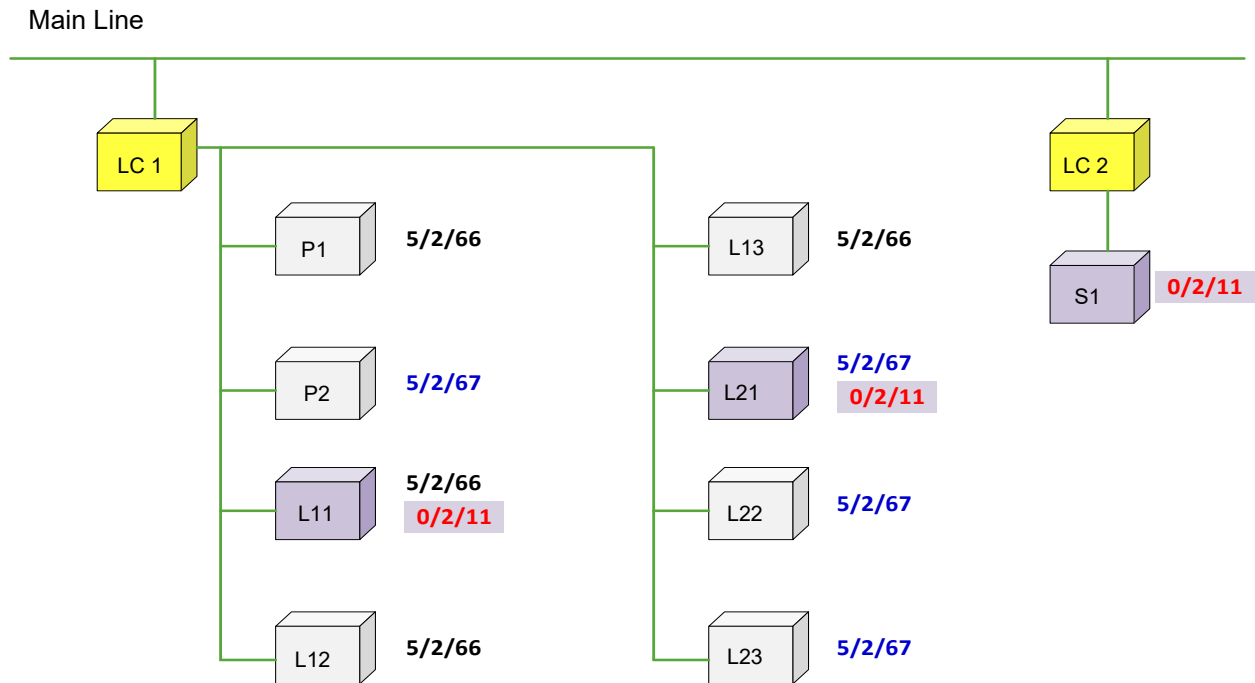


Figure 13: Line-crossing telegram

The brightness sensor (S1) is in line 2, but it needs to control lights L11 and L21 in line 1. Since S1 is not in the same line as the lights, its telegram must travel through the main line.

Step-by-step process:

1. S1 sends a telegram with group address 0/2/11.
2. Line coupler LC 2 detects that bus devices in another line (L1) respond to this telegram and forward it to the main line.
3. Line coupler LC 1 detects that devices in line 1 need to process group address 0/2/11 and forwards it into line 1.
4. All bus devices in line 1 receive the telegram, but only the actuators for L11 and L21 execute the command.

13 Area-crossing telegram

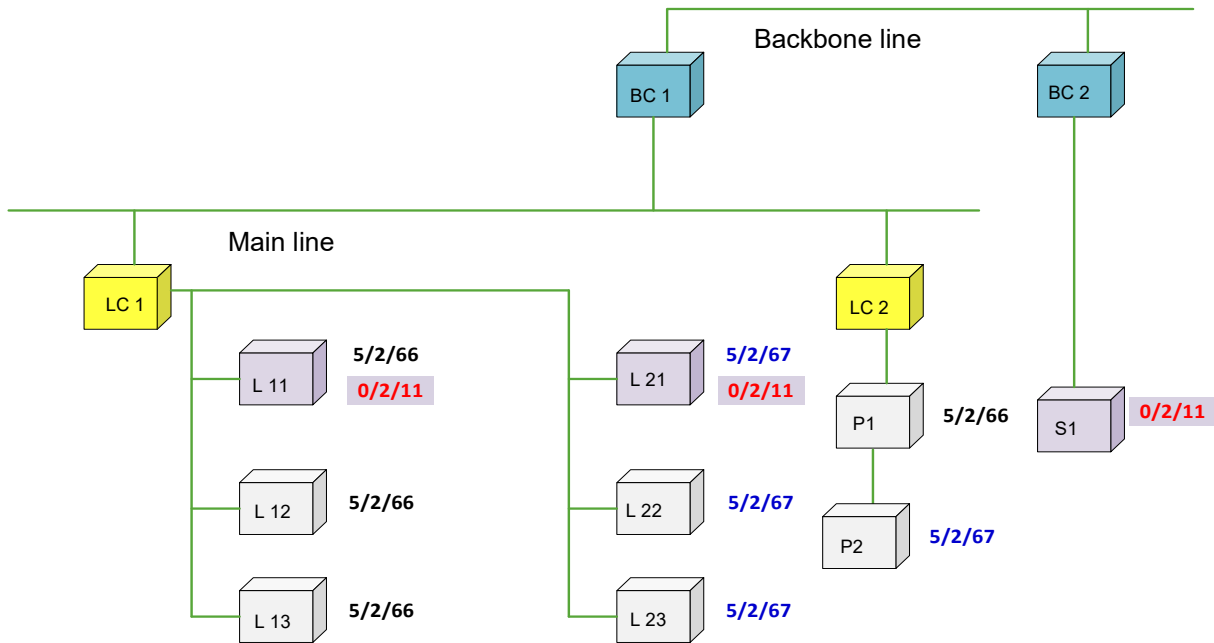


Figure 14: Area-crossing telegram

The brightness sensor (S1) is located in a different area, but it still needs to control lights L11 and L21 in area 1, line 1. Since S1 is outside area 1, its telegram must travel through the backbone line.

Step-by-step process:

1. S1 sends a telegram with group address 0/2/11.
2. Backbone coupler BC 2 detects that bus devices outside its own area respond to this telegram and forwards it onto the backbone line.
3. Backbone coupler BC 1 recognizes that devices in area 1 need the telegram and routes it to line coupler LC 1.
4. Line coupler LC 1 forwards the telegram into line 1.
5. All bus devices in line 1 receive the telegram, but only the actuators for L11 and L21 execute the command.

14 Hop count in KNX couplers

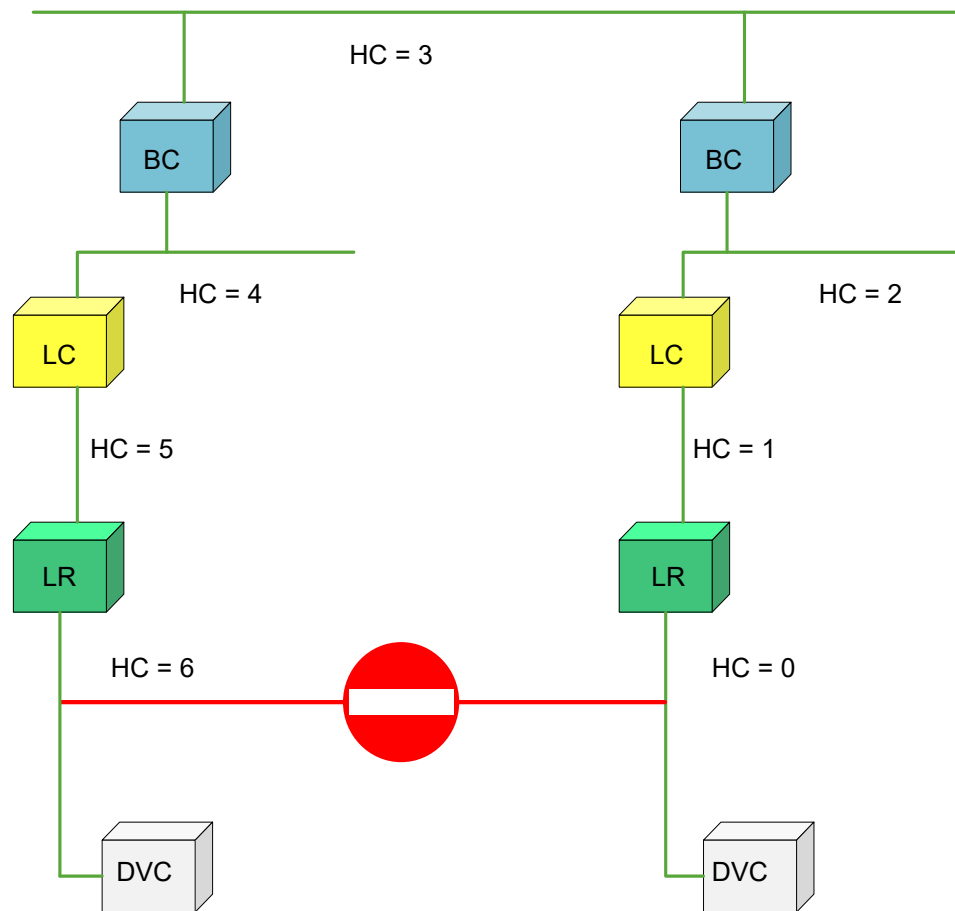


Figure 15: Hop count

The Hop count is a mechanism in KNX TP that controls how far a telegram can travel across different lines and areas. It prevents telegrams from endlessly circulating in the network, especially in cases of accidental loops.

How the hop count works:

- Every telegram sent by a device starts with a **hop count value of 6**.
- Each line coupler, backbone coupler, or line repeater decreases the hop count by 1 before forwarding the telegram.
- The telegram continues to be passed on until the hop count reaches 0.
- Once the counter reaches 0, the telegram is no longer forwarded, ensuring efficient network operation.
- Apart from that, the coupler only forwards telegrams if the filter table allows it. This additionally prevents unnecessary telegram traffic.

15 TP Topology – Building structure

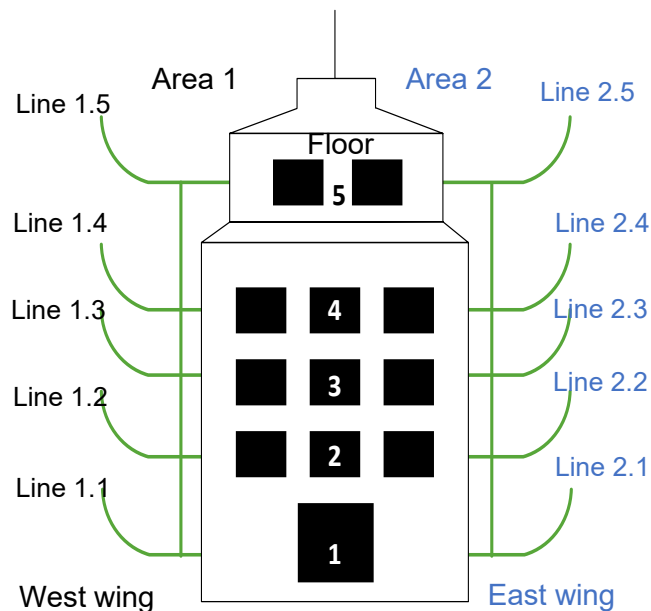


Figure 16: Division of lines in a medium sized project (example)

Area		Line	
Number	Comment	No.	Comment
0	Area 0	0	Backbone line
1	West wing	0	Main line West
		1	1st floor
		2	2nd floor
		3	3rd floor
		4	4th floor
		5	5th floor
2	East wing	0	Main line East
		1	1st floor
		2	2nd floor
		3	3rd floor
		4	4th floor
		5	5th floor

When designing a KNX TP installation in a medium-sized building, it is best to align the topology with the building's physical layout. This makes the system easier to install, manage, and troubleshoot.

- Line numbers = Floor numbers
 - Example:
 - Line 1: First floor
 - Line 2: Second floor
- Area numbers = Building or wing numbers
 - Example:
 - Area 1: West wing of building
 - Area 2: East wing of the same building

Although this ideal setup is not always feasible, designing the topology based on the building's physical layout simplifies installation. It increases reliability due to the function and performance of line couplers and the requirement to have a KNX power supply on each line, the main line, and the backbone line.

16 Backbone- /line coupler classical structure

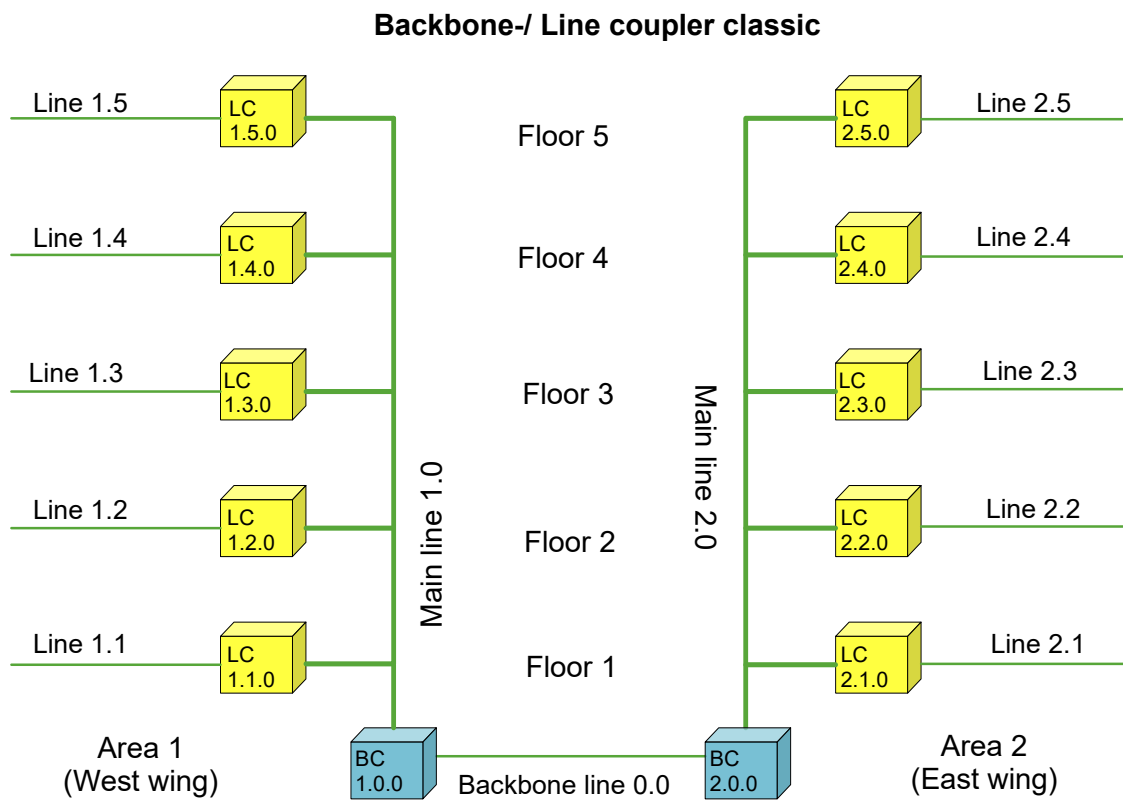


Figure 17: Backbone- /line coupler classical structure

17 KNX internal and external interfaces

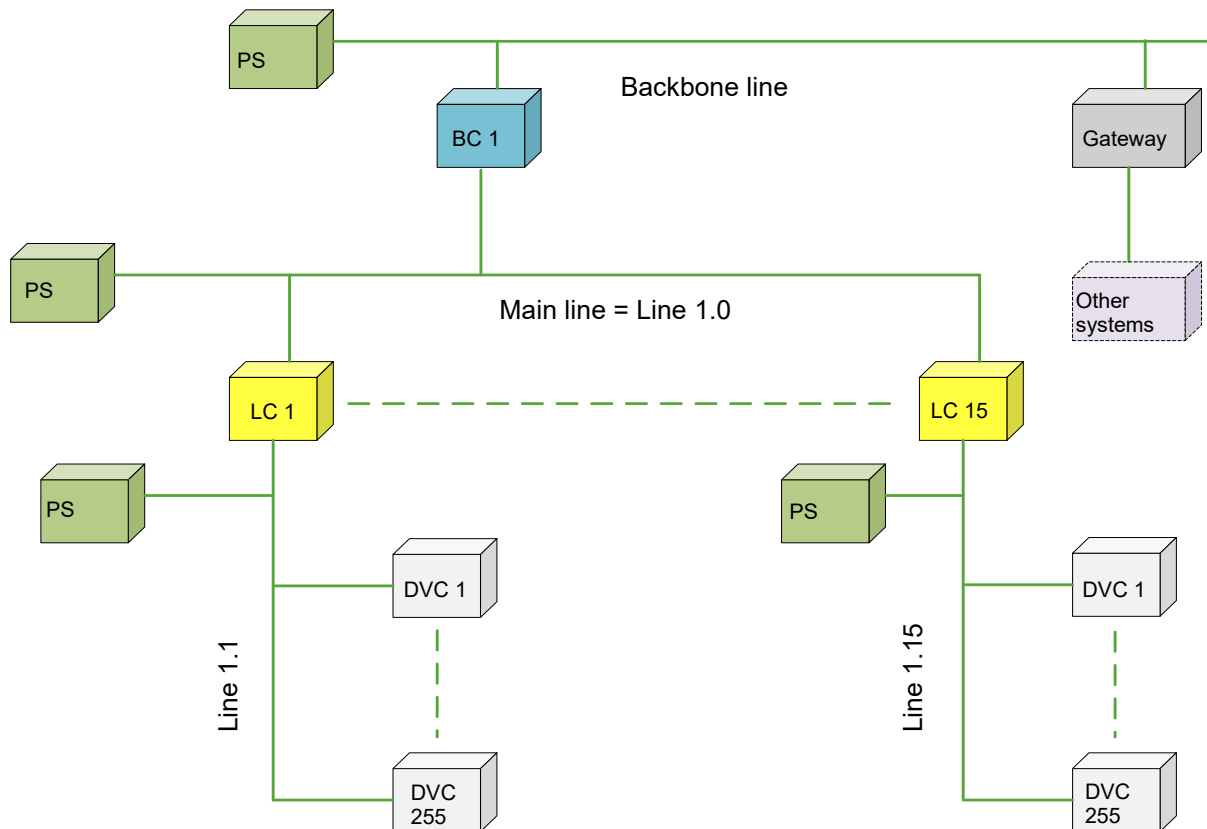


Figure 18: KNX - internal and external interfaces

KNX is highly flexible and can be integrated with other systems using various interfaces and gateways:

- The backbone line (or any other line) can be connected to external systems using a gateway.
- Gateways allow bidirectional protocol conversion, enabling communication between KNX and other technologies.
- Examples of external systems that can connect to KNX:
 - AV Systems (Audio/Video automation)
 - Logic Controllers (for advanced automation)
 - Building Management Systems (BMS)
 - Visualization platforms (Smart Home dashboards)
 - Internet & Remote Access (for cloud-based control)
- Gateways should be placed at the highest level in the KNX topology (e.g., backbone line) to:
 - Minimize traffic on each lower line.
 - Ensure messages reach external interfaces efficiently.

In a similar way KNX media couplers are used to connect different types of KNX media (e.g., TP and RF or IP and TP). It's also possible to connect parts of the KNX installation via optical fiber which allows long-distance communication with KNX installations. The key advantages of optical fiber:

- Greater cable lengths: Covers longer distances than TP.
- Electrical isolation: Prevents interference and grounding issues.
- High-speed and secure communication: Ideal for large building automation projects.

18 KNX IP network for higher telegram rates

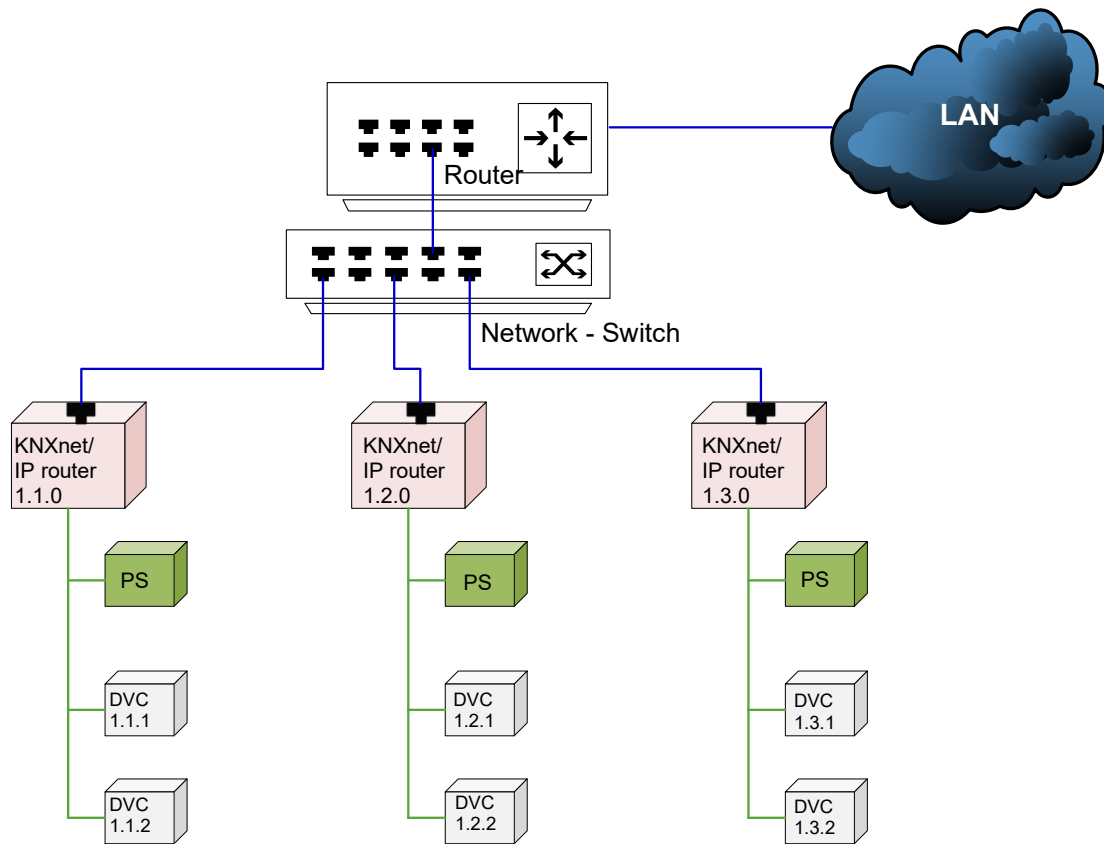


Figure 19: IP Network: higher telegram rates

As KNX installations grow and integrate with external systems, the telegram traffic increases, which can overload a pure TP-based topology. To handle higher data rates and improve communication speed, the TP main or backbone line can be replaced by IP. Line couplers and backbone couplers are then replaced by KNXnet/IP routers.

Why use IP and not TP in the main or backbone line?

- Traditional TP networks are limited to 9,600 bits per second which may not be enough for large projects.
- IP networks are significantly faster – using Gigabit switches, telegrams can be transmitted 100,000 times faster than TP.
- KNX over IP allows integration with external systems for centralized control, visualization, and remote access.

KNX Tunneling & Routing via KNXnet/IP:

- KNXnet/IP routers (sometimes referred to as KNX IP router) use Tunneling and Routing for IP-based communication:
 - **Tunneling:** Used for ETS programming over IP.
 - **Routing:** Works like a line coupler, sending and receiving telegrams across lines taking into account the filter table.
- How routing works over IP:
 - A KNXnet/IP router sending a line-crossing telegram will transmit this telegram with a Multicast IP address.
 - All other KNXnet/IP routers in the network listen to multicast messages and check their filter tables before forwarding or blocking the telegram.

Important notes about multicast in KNX IP networks:

- KNX uses a dedicated worldwide registered multicast address (**224.0.23.12**), pre-configured in KNXnet/IP routers.
- This multicast address can be changed in ETS within the allowed IP address range.
- The network switch or IP router in the LAN must support multicast traffic. Always consult the network administrator before implementation.
- Multicast addresses do not work across the internet, except over a VPN connection.

Differences between KNX IP Interfaces and KNXnet/IP routers:

Feature	KNX IP Interface	KNXnet/IP router
Tunneling (ETS Programming)	Yes	Yes
Routing (telegram filtering & forwarding)	No	Yes
Multicast support for line-crossing Telegrams	No	Yes

- **KNX IP interfaces:** Used for ETS programming and remote access, but do not support routing.
- **KNXnet/IP routers:** They support both Tunneling and Routing, making them ideal for replacing TP line and backbone couplers, but also serve as interfaces for remote access.

Remote access & cloud-based control:

Some KNX devices offer **secure** remote access via a cloud portal, allowing:

- Remote programming via ETS (Tunneling).
- Monitoring and control of group communication.
- Service and maintenance from anywhere.

However, these remote access devices connect as IP devices and do not offer routing functions.

19 Line couplers replaced by KNXnet/IP routers

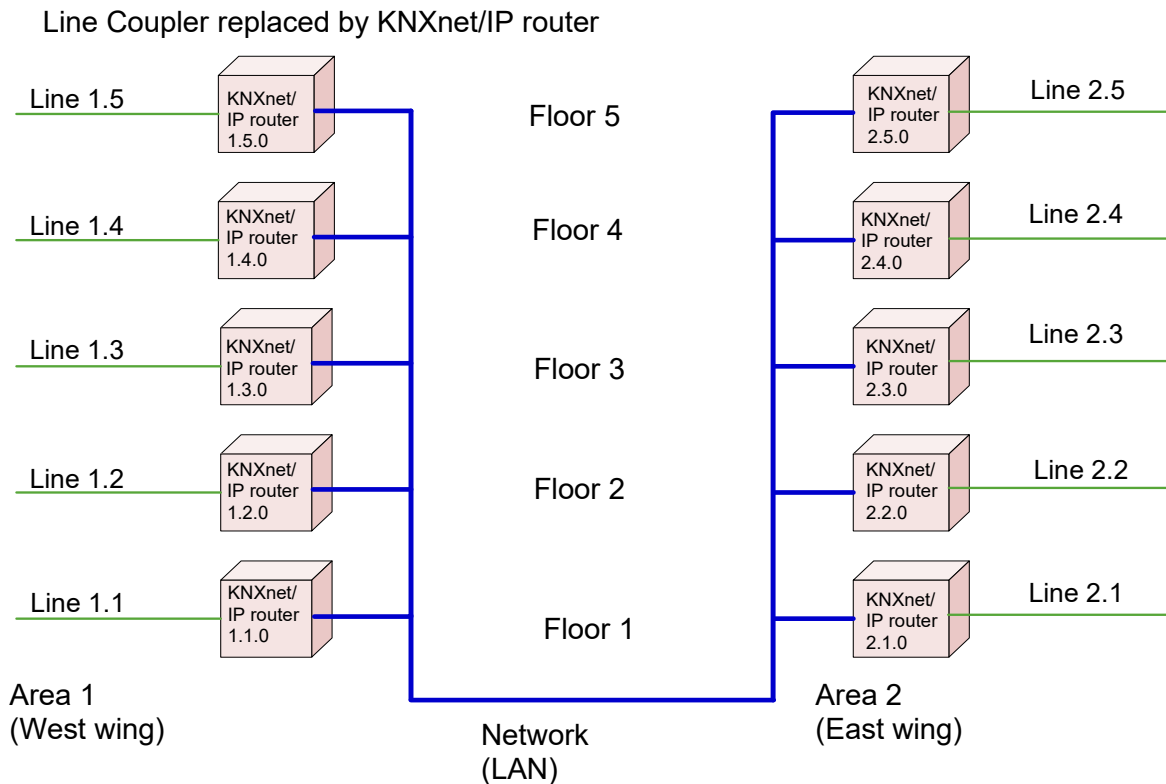


Figure 20: Our picture again: Line couplers have been replaced by KNXnet/IP routers. This picture represents the underneath explained case 1.

KNXnet/IP routers can replace traditional line couplers and backbone couplers, improving communication speed by using Ethernet instead of TP for main and backbone lines.

Two options for replacing KNX TP Couplers

Case 1: Line couplers replaced by KNXnet/IP routers:

- Main lines and backbone lines are fully replaced by Ethernet.
- Each KNX TP line remains unchanged but connects via a KNXnet/IP router instead of a line coupler.
- This significantly increases communication speed and reduces telegram congestion.

Case 2: Only backbone couplers replaced by KNXnet/IP routers

- Main lines remain TP-based, with traditional line couplers in place.
- Only the backbone line is replaced by Ethernet.
- This improves backbone speed while keeping existing TP main lines.

Which case is better?

- **Case 1** (Fully Ethernet-based backbone and main lines): Best for high telegram-rate projects.
- **Case 2** (Ethernet backbone, TP main lines): Best for existing TP installations needing faster backbone communication.
- **A combination of both cases** is possible but should only be used in exceptional situations (covered in the KNX Advanced Course).

20 Limits to using KNXnet/IP routers

The Ethernet allows high telegram rates and minimizes telegram losses. But it cannot prevent traffic overload if telegrams from multiple lines are funneled into a single line.

Metaphor:

Imagine a 1,000-lane highway where cars enter through 100 different entry points, but all need to exit through a single lane - this creates a massive traffic jam.

- The same happens when many telegrams from different lines are sent into a single line simultaneously.
- This can lead to bottlenecks, delays, or even telegram loss.

This is not just a KNX issue - it's a common challenge in all mesh-structured networks.

How to prevent data overload in KNXnet/IP networks?

- By proper device placement in the topology.
- By setting up devices logically to avoid excessive telegram traffic on a single line.
- By using filter tables in KNXnet/IP routers.
- By configuring the filter tables in coupler devices (KNXnet/IP routers) to limit unnecessary telegrams from crossing lines.
- By managing feedback telegrams and cyclical transmissions in a manner appropriate to the situation.
- By limiting automatic status updates (feedback telegrams), especially when many devices report to a single system.
- By being cautious with the repetition rate of cyclical telegrams (e.g., from sensors that send data at fixed intervals) to avoid overloading the network.